

HOUSE PRICE PREDICTION USING MACHINE LEARNING

Student Details:

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Task Number: Task 2 – House Price Prediction

Objective:

The objective of this project is to build a machine learning model that predicts house prices using various housing features such as living area, bedrooms, bathrooms, floors, location details, and property conditions. The project aims to estimate house prices accurately using regression techniques.

Dataset:

The House Price dataset contains various housing attributes including:

- Date of sale
- Number of bedrooms
- Number of bathrooms
- Living area (sqft_living)
- Lot area (sqft_lot)
- Number of floors
- Waterfront information
- House view and condition
- Basement area
- Year built and year renovated
- Street, city, state, and country details

The target variable is:

- House Price

Exploratory Data Analysis:

Data analysis was performed using Pandas, Matplotlib, and Seaborn. The dataset was examined for missing values, data types, and statistical information. Various visualization techniques such as scatter plots, histograms, box plots, and pair plots were used to understand the relationships between features and house prices.

Data Preprocessing:

The categorical features were converted into numerical values using Label Encoding. The dataset was divided into training and testing datasets using an 80:20 ratio. The data preprocessing stage helped improve the performance of machine learning models.

Models Used:

The following machine learning algorithms were implemented:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

The models were trained and evaluated to determine the best-performing algorithm for house price prediction.

Results:

The models were evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Linear Regression produced satisfactory results, while Random Forest achieved better prediction accuracy. The trained model was saved as `house_model.pkl` for future predictions.

A sample prediction was performed successfully, and the model was able to estimate house prices based on housing features.

Tools and Technologies:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Scikit-learn
- Joblib

Conclusion:

This project successfully demonstrated the complete machine learning workflow including data collection, preprocessing, data visualization, model training, evaluation, prediction, and model saving. The developed model can be used to estimate house prices and provides practical experience in regression analysis and predictive modeling using machine learning techniques.